CS4865/6865 Fall 2025

Laboratory #1

Friday, September 26, 9:30am in GC112 (in person)

Introduction to network simulation I

- packet transmission time calculations

- network traffic visualization with NS2 simulation and network animator NAM

- determining packet transmission time from the NS2 simulation trace file

Reference materials:

1. Mark Greis, Tutorial for the Network Simulator "ns'
2. TCL description.
3. ns Manual.
4. Trace file format: https://www.isi.edu/websites/nsnam/ns/doc/node289.html
5. nam trace file format: https://www.isi.edu/websites/nsnam/ns/doc/node616.html

General instructions:

- ns2 simulator runs under Linux in our FCS labs.
- If you are accessing the lab remotely please test the setup BEFORE the lab session:
  - Complete and test the setup required for remote access to the FCS labs with GUI: https://www.cs.unb.ca/help/remote-lab-gui-access.shtml
- Complete lab exercises and prepare the lab report.
- During the lab (labs ONLY, not assignments) you may work in groups, however, indi D2L submissions are required from each student.

Exercise 1

1.1 Review the webpage on ns2 simulation prepared by Prof. DeDourek (no writeup necessary

http://www.cs.unb.ca/~dedourek/ns2.html

Exercise 2

2.1 Complete the simulation described in Section IV of Mark Greis' tutorial. Create a work dir for Lab 1, then create a file with the simulation code (enter the TCL code using a text edit the entire file from the tutorial site). Run the ns2 simulator from the command window:

```
ns name_of_the_tcl_file.tcl
```

Submit the tcl file (embed the text in your lab report) and a snapshot of the simulator scree showing the simulation run (i.e. nam screenshoot).

Exercise 3

3.1 Show the first thirty lines of the trace file out.nam
3.2 Show the line in the trace file corresponding to the first packet being transmitted.
3.3 Show the line in the trace file corresponding to the first packet being received.

Exercise 4

4.1 Complete simulation of a packet forwarding network consisting of three nodes: the sender, forwarding node and the receiver. Use the same link parameters and the packet size as in E 2.

Submit the tcl file (embed the text in your lab report) and a snapshot of the simulator scree showing the simulation run.

Exercise 5

5.1 Show the first thirty lines of the trace file out.nam from the simulation in Exercise 4

5.2 - Show the line in the trace file corresponding to the first packet being transmitted. What's stamp?

- Show the line in the trace file corresponding to the first packet being received at the forw node. What's the time stamp?

- Show the line in the trace file corresponding to the first packet being received at the desti node. What's the time stamp?

- How long does it take to deliver the first packet to its destination? Can it be calculated fro timestamps recorded in the trace file (show how)?

Exercise 6

6.1 Can the delay of the first packet (from Exercise 5) be calculated theoretically, i.e. derived f network topology, link data rates, link delays and packet size? Show the formula and calcu delay. Is this value the same as what you calculated in Exercise 5.2?

Complete and then prepare the documentation for Exercises 1 to 6 (use the headings indicating the question number) and store it in a **single PDF file**. Include a preamble (title, section) with your na student number, course number etc. Please number all the pages! Submit this single PDF file on D Lab 1 Report by the due date indicated in the D2L dropbox.

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